

Specification of MEMS Microphone (RoHS Compliance & Halogen Free)

Customer Name :

Customer Model :

Goertek Model : S15OT421-005

Goertek			CUSTOMER APPROVAL
<u>DESIGN</u>	<u>Jasen</u>	<u>2019.06.04</u>	
<u>CHKD</u>	<u>Samual</u>	<u>2019.06.04</u>	
<u>STANDARD</u>	<u>Sweety</u>	<u>2019.06.04</u>	
<u>APVD</u>	<u>Daniel</u>	<u>2019.06.04</u>	

Tel : + 86 536 3051234

E- Mail : goertek@goertek.com

Website: <http://www.goertek.com>

Address: No.268 Dongfang Road, High-Tech Industry
Development District, Weifang, Shandong, P.R.C.

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1 Security Warning

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2 Publication History

Version	Date	Description	Author	Approved
1.0	2019.06.04	New Design	Jasen	Daniel

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1 Introduction:

MEMS MIC which is able to endure reflow temperature up to 260 °C for 50 seconds can be used in SMT process. It is widely used in telecommunication and electronics device such as mobile phone, MP3, PDAs etc.

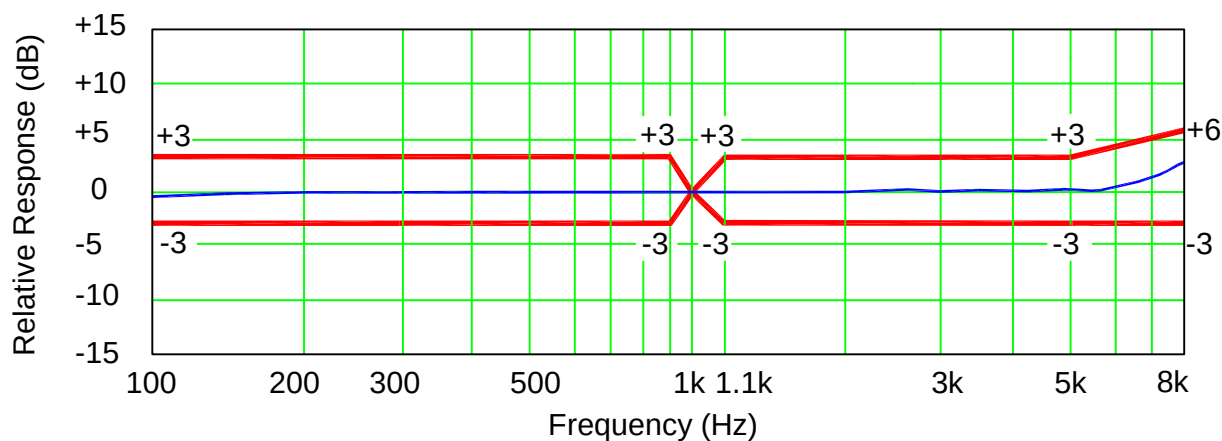
2 Test Condition (Vs=2.0V,L=50cm)

Standard Conditions (As IEC 60268-4)	Temperature	Humidity	Air pressure
Environment Conditions	+15°C ~ +35°C	25%RH ~ 75%RH	86kPa ~ 106kPa
Basic Test Conditions	+20±2°C	60%RH ~ 70%RH	86kPa ~ 106kPa

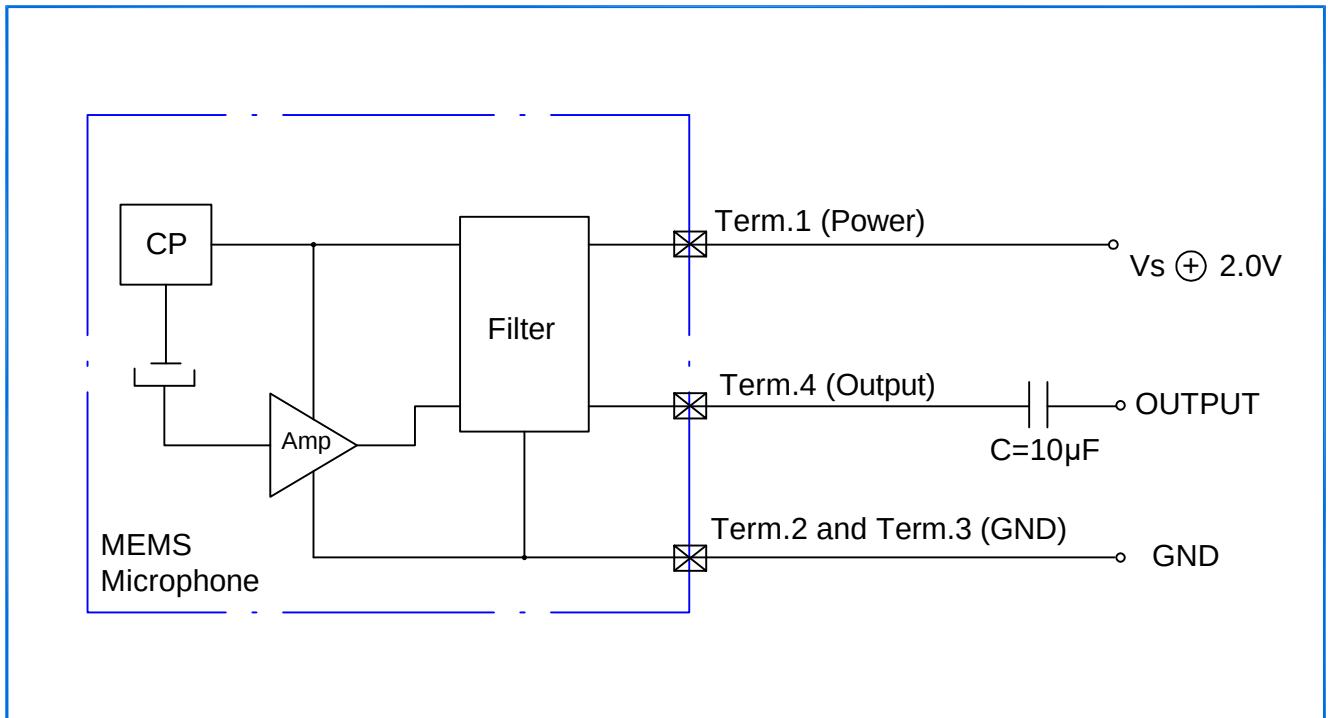
3 Electrical Characteristics

Item	Symbol	Test Conditions	Min	Typ	Max	Unit
Sensitivity	S	f=1kHz, Pin=1Pa	-43	-42	-41	dB
Output Impedance	Zout	f=1kHz, Pin=1Pa			400	Ω
Directivity	D(θ)		Omnidirectional			
Current Consumption	I		50		150	μA
S/N Ratio	S/N(A)	f=1kHz, Pin=1Pa A-Weighted		59		dB
Decreasing Voltage Characteristic	ΔS	f=1kHz, Pin=1Pa Vs=3.6 ~ 1.5V	No Change			
Operating Voltage Range	Vs		1.5		3.6	V
Total Harmonic Distortion	THD	110dB SPL@ f=1kHz			1	%

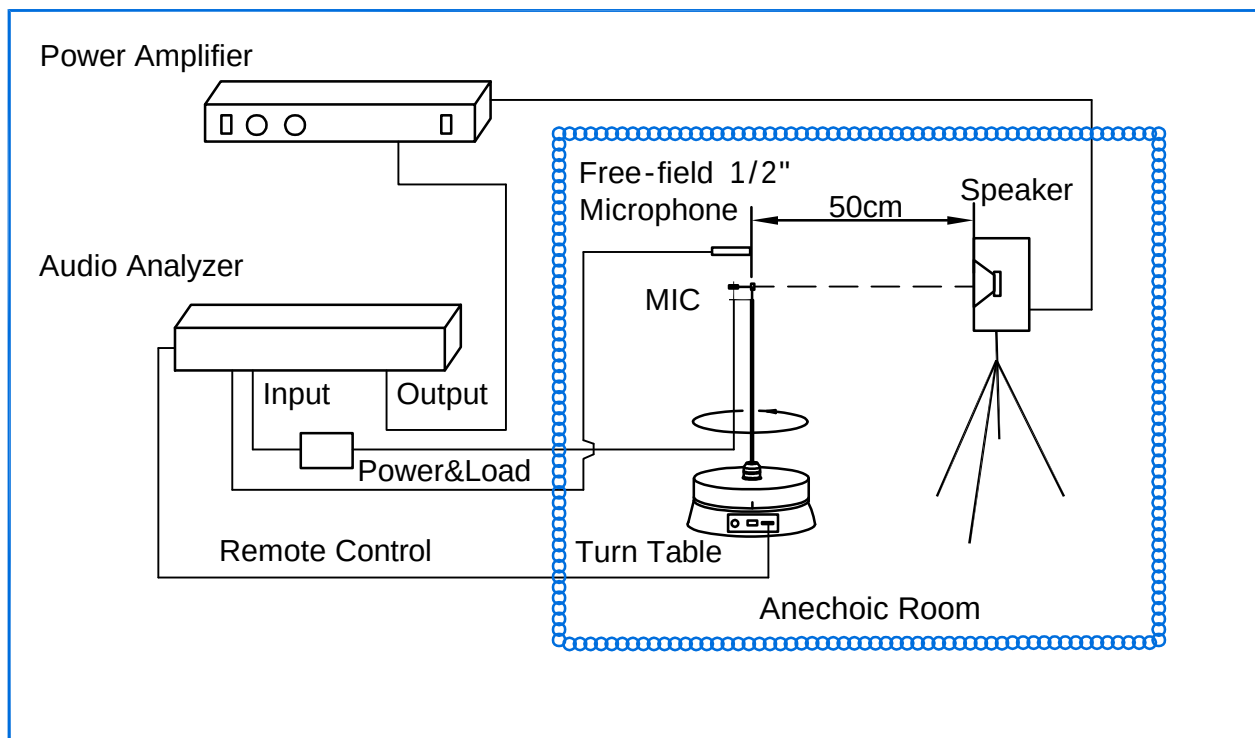
4 Frequency Response Curve Limits



5 Measurement Circuit

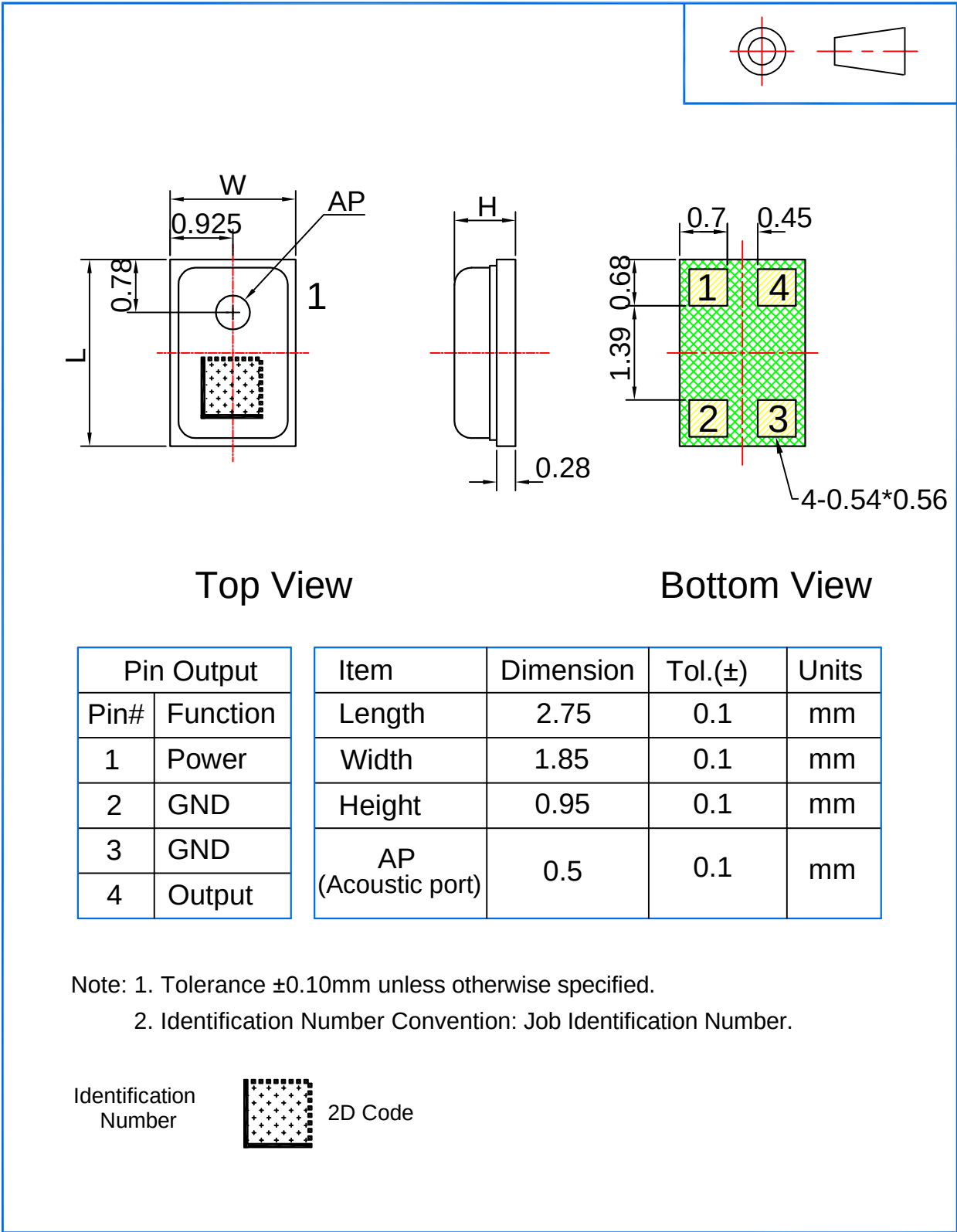


6 Test Setup Drawing



7 Mechanical Characteristics

7.1 Appearance Drawing (Unit: mm)



7.2 Weight

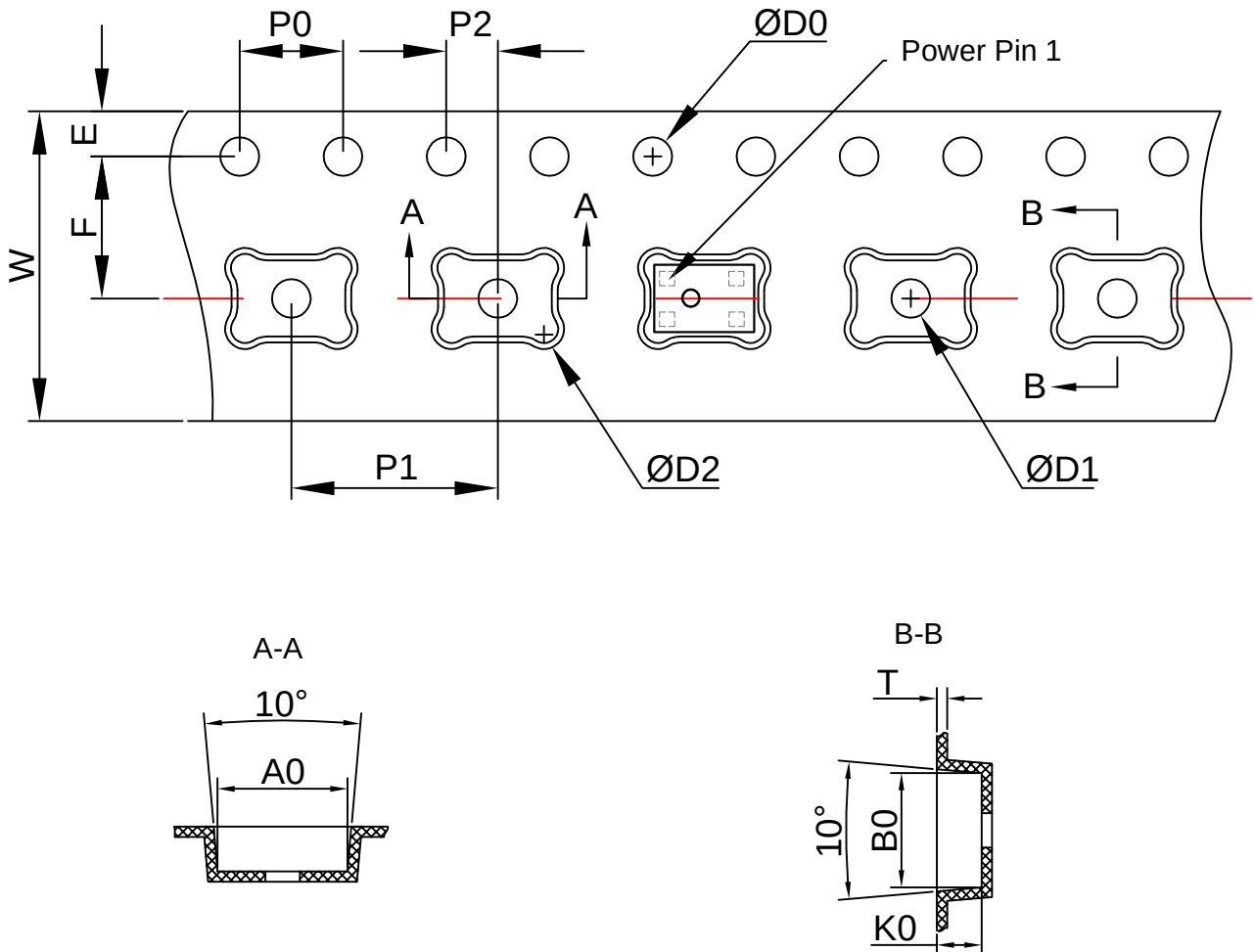
The weight of the MIC is Less than 0.04g.

8 Reliability

8.1 Vibration Test	To be no interference in operation after vibrations, 4 cycles, from 20 to 2,000Hz in each direction(X,Y,Z), 48 minutes, using peak acceleration of 20g, sensitivity should vary within $\pm 3\text{dB}$ from initial sensitivity. (The measurement to be done after 2 hours of conditioning at $+15^{\circ}\text{C} \sim +35^{\circ}\text{C}$, R.H 25%~75%)
8.2 Drop Test	To be no interference in operation after dropped to 1.0cm steel plate 18 times from 1.5 meter height, sensitivity should vary within $\pm 3\text{dB}$ from initial sensitivity. (The measurement to be done after 2 hours of conditioning at $+15^{\circ}\text{C} \sim +35^{\circ}\text{C}$, R.H 25%~75%)
8.3 Temperature Test	a) After exposure at $+125^{\circ}\text{C}$ for 200 hours, sensitivity should vary within $\pm 3\text{dB}$ from initial sensitivity. (The measurement to be done after 2 hours of conditioning at $+15^{\circ}\text{C} \sim +35^{\circ}\text{C}$, R.H 25%~75%) b) After exposure at -40°C for 200 hours, sensitivity should vary within $\pm 3\text{dB}$ from initial sensitivity. (The measurement to be done after 2 hours of conditioning at $+15^{\circ}\text{C} \sim +35^{\circ}\text{C}$, R.H 25%~75%)
8.4 Humidity Test	After exposure at $+85^{\circ}\text{C}$ and 85% relative humidity for 200 hours, sensitivity should vary within $\pm 3\text{dB}$ from initial sensitivity. (The measurement to be done after 2 hours of conditioning at $+15^{\circ}\text{C} \sim +35^{\circ}\text{C}$, R.H 25%~75%)
8.5 Mechanical Shock Test	Then subject samples to three one-half sine shock pulses (3000 g for 0.3 milliseconds) in each direction (for six axes in total) along each of the three mutually perpendicular axes for a total of 18 shocks, sensitivity should vary within $\pm 3\text{dB}$ from initial sensitivity. (The measurement to be done after 2 hours of conditioning at $+15^{\circ}\text{C} \sim +35^{\circ}\text{C}$, R.H 25%~75%)
8.6 Thermal Shock Test	After exposure at -40°C for 30 minutes, at $+125^{\circ}\text{C}$ for 30 minutes (change time 20 seconds) 5 cycles, sensitivity should vary within $\pm 3\text{dB}$ from initial sensitivity. (The measurement to be done after 2 hours of conditioning at $+15^{\circ}\text{C} \sim +35^{\circ}\text{C}$, R.H 25%~75%)
8.7 Reflow Test	Adopt the reflow curve of item 12.3, after five reflows, sensitivity should vary within $\pm 2\text{dB}$ from initial sensitivity. (The measurement to be done after 2 hours of conditioning at $+15^{\circ}\text{C} \sim +35^{\circ}\text{C}$, R.H 25%~75%)
8.8 Electrostatic Discharge Test	Under $C=150\text{pF}$, $R=330\text{ohm}$. Tested to $\pm 8\text{KV}$ contact to the case and tested to $\pm 2\text{KV}$ contact to I/O terminals.10 times. Grounding. Sensitivity should vary within $\pm 2\text{dB}$ from initial sensitivity.

9 Package

9.1 Tape Specification



The Dimensions as Follows:

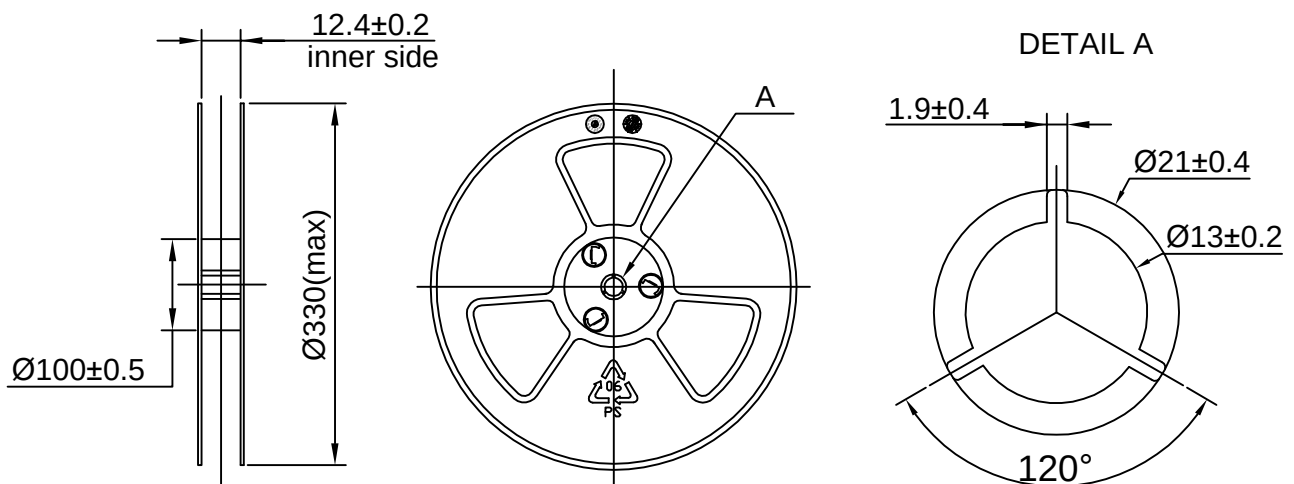
ITEM	W	E	F	$\varnothing D0$	$\varnothing D1$
DIM(mm)	12.0±0.30	1.75±0.10	5.5±0.05	1.50 ^{+0.10} ₀	1.00 ^{+0.10} ₀
ITEM	P0	10P0	P1	A0	B0
DIM(mm)	4.00±0.10	40.00±0.20	8.00±0.10	3.00±0.05	2.05±0.05
ITEM	K0	P2	T	$\varnothing D2$	
DIM(mm)	1.10±0.10	2.00±0.1	0.30±0.05	0.50±0.10	

9.2 Reel Dimension

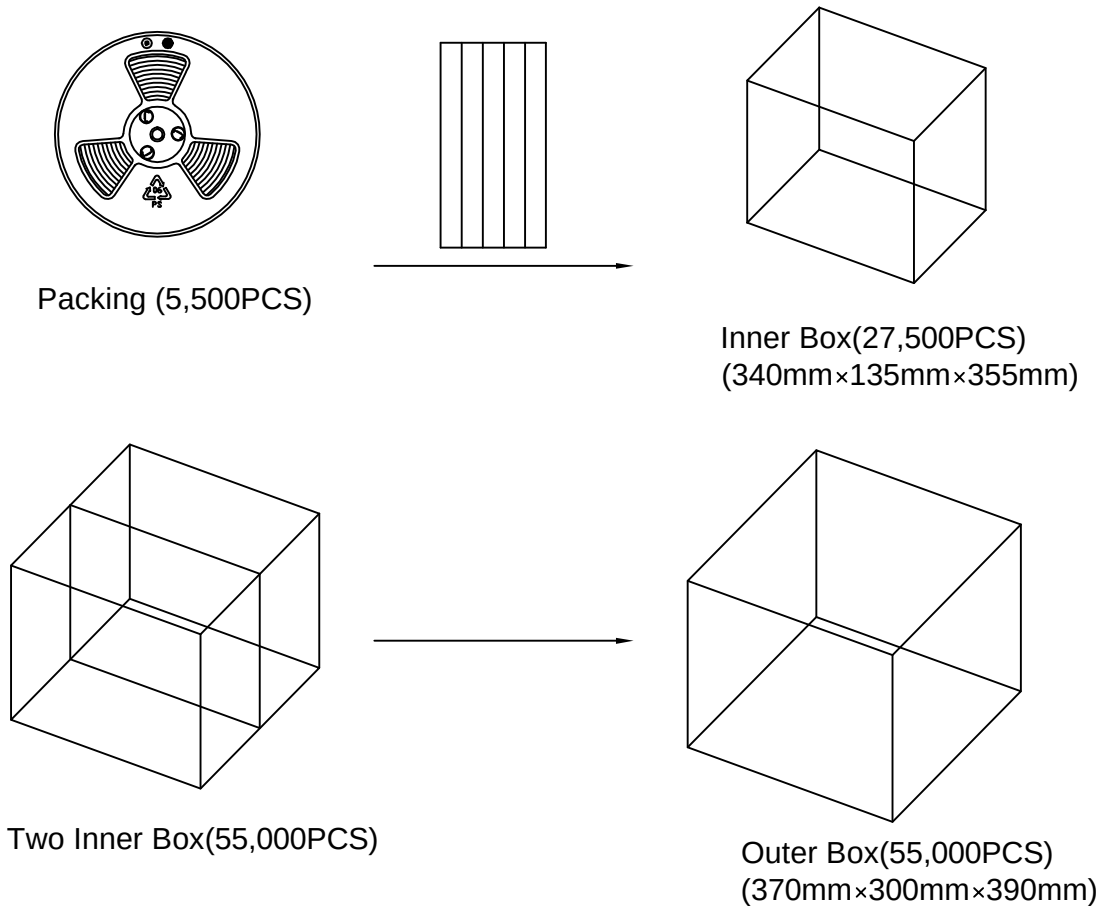
7" reel for sample stage

13" reel will be provided for the mass production stage

The following is 13" reel dimensions (unit:mm)

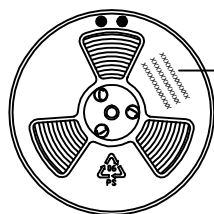


9.3 The Content of Box(13" reel)



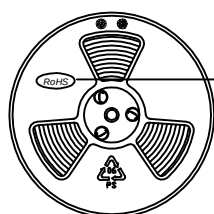
9.4 Packing Explain

9.4.1 The Label Content of the Reel



The Content Includes:
Product type, Lot, Customer P/N;
and other essential information such as
Quantity, Date etc.

9.4.2 The RoHS Label



RoHS HF
Compliance Mark

10 Storage and Transportation

10.1 Keep MEMS MIC in warehouse with less than 75% humidity and without sudden temperature change, acid air, any other harmful air or strong magnetic field. Recommend storage period no more than 1 year and floor life(out of bag) at factory no more than 4 weeks.

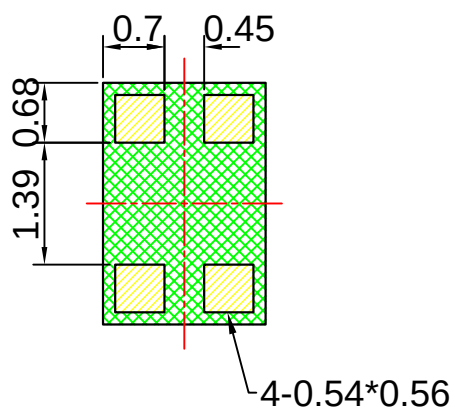
10.2 The MEMS MIC with normal pack can be transported by ordinary conveyances. Please protect products against moist, shock, sunburn and pressure during transportation.

10.3 Storage Temperature Range : $-40^{\circ}\text{C} \sim +70^{\circ}\text{C}$ (Microphone units with package)

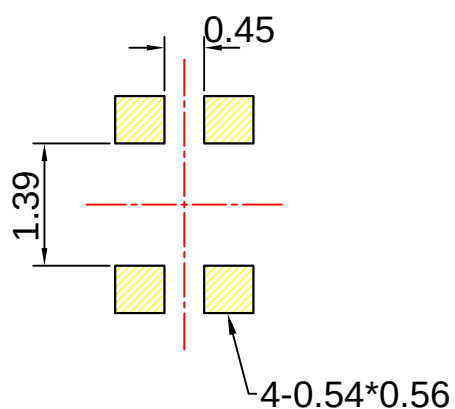
10.4 Operating Temperature Range : $-40^{\circ}\text{C} \sim +100^{\circ}\text{C}$

11 Land Pattern Recommendation

11.1 The Pattern of MIC Pad(Unit:mm)



11.2 Recommended Soldering Surface Land Pattern

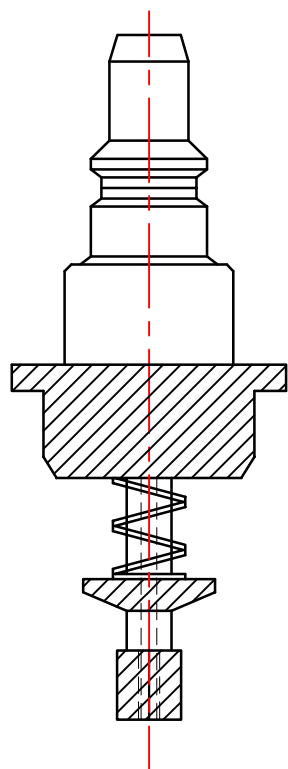


12 Soldering Recommendation

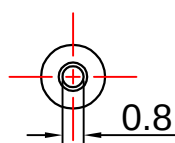
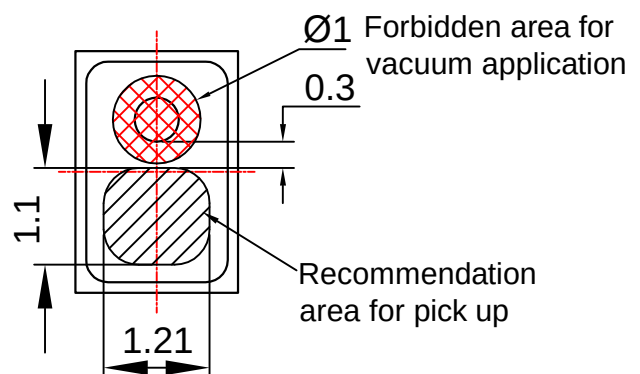
12.1 Soldering Machine Condition

Temperature Control	8 zones
Heater Type	Hot Air
Solder Type	Lead-free

12.2 The Drawing and Dimension of Nozzle

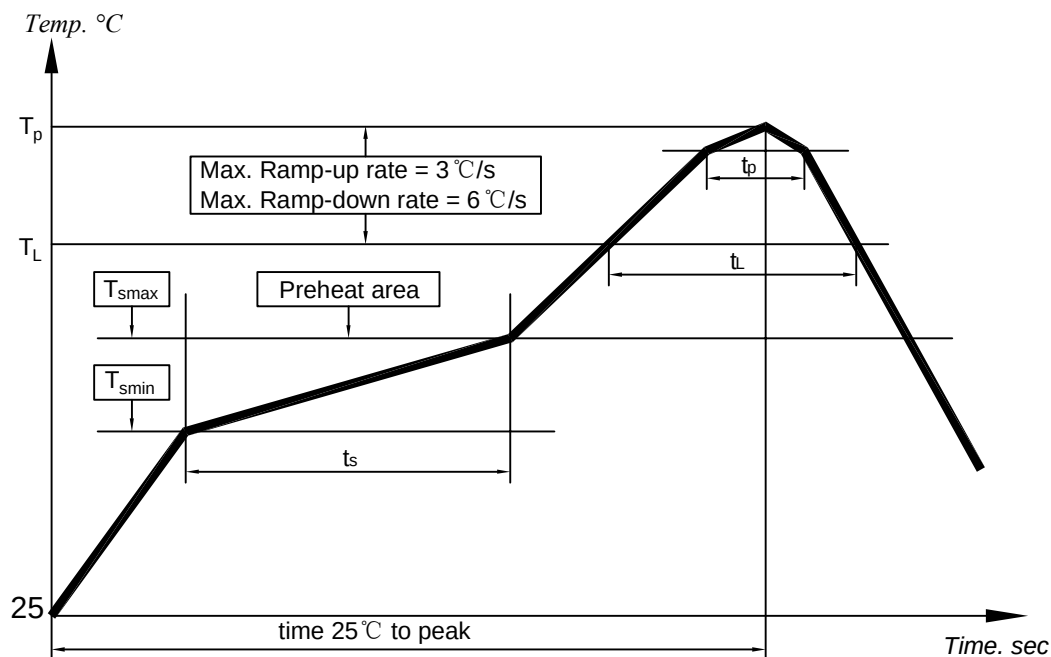


Nozzle Diameter: 0.8mm



Please don't vacuum over the acoustic port directly. The recommendation is for reference.

12.3 Reflow Profile



Key Features of The Profile:

Average Ramp-up rate(T_{smax} to T_p)	3°C/s max.
Preheat : Temperature Min(T_{smin}) Temperature Max(T_{smax}) Time(T_{smin} to T_{smax})(t_s)	150°C 200°C 60~180s
Time maintained above : Temperature(T_L) Time(t_L)	217°C 60~150s
Peak Temperature(T_p)	260°C
Time within 5°C of actual Peak Temperature(t_p) :	30~40s
Ramp-down rate(T_p to T_{smax})	6°C/s max
Time 25°C to Peak Temperature	8min max

When MEMS MIC is soldered on PCB, the reflow profile is set according to solder paste and the thickness of PCB etc.

12.4 Rework

- (1) 250°C~270°C, maximum 30 sec, Peak temperature 330°C.
- (2) Wind speed: 15L/m.
- (3) It is very important not to put a heatgun over the acoustic port of the microphone.

13 Cautions

13.1 Board Wash Restrictions

It is very important not to wash the PCBA after reflow process, otherwise this could damage the microphone.

13.2 Nozzle Restrictions

It is very important not to be put a nozzle over the acoustic hole of the microphone, otherwise this could damage the microphone.

13.3 Blowing Restrictions

It is very important not to blow the acoustic port of the microphone directly, otherwise this could damage the microphone.

13.4 Ultrasonic Restrictions

It is very important not to use ultrasonic process. otherwise this could damage the microphone.

13.5 Case Adaption to Pressure Restrictions

It is very important not to press the case with a force larger than 2.5kgf, otherwise this would damage the microphone.

14 Output Inspection Standard

Output inspection standard is executed according to <<ISO2859-1:1999>>.